# Why Proprietary Reasoning Models Lock You In

How open-weight test-time compute brings verifiable mathematical reasoning to consumer GPUs.

## THE REASONING PAYWALL

Closed frontier reasoning models deliver impressive benchmarks on math and code, but they lock your infrastructure into high per-token pricing and hide internal reasoning traces inside black boxes.

```
Closed APIs hide internal thinking steps, preventing error audit
```

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github.com/deepseek-ai/DeepSeek-R1 →

# Distilling Test-Time Compute into Qwen

Architectural Shift

ARCHITECTURAL SHIFT

DeepSeek distilled 800k verifiable reasoning trajectories from the 671B frontier model directly into standard Qwen 2.5 architectures (1.5B, 7B, 14B, 32B), transferring long-chain logic patterns.

```
Distilled Checkpoints: 1.5B, 7B, 14B, 32B under MIT License
```

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# Runs on 8GB to 16GB Consumer GPUs

Consumer Hardware

CONSUMER HARDWARE

The 7B and 14B checkpoints run smoothly in 4-bit and 8-bit quantization using Ollama, LM Studio, or vLLM on a single RTX 3060/4070 or Apple Silicon Mac, costing $0 in cloud API fees.

```
Memory footprint: ~5GB VRAM for 7B INT4 • $0 Cloud Fees
```

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# Thinking Tokens Multiply Total Latency

The Latency Catch

THE LATENCY CATCH

R1-Distill models can output 2,000 to 6,000 thinking tokens inside tags before producing an answer. While accuracy is outstanding, total generation time is 3x to 5x longer than standard instruction models.

```
Trade-off: High reasoning depth vs real-time chat latency
```

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# Auditable White-Box Agent Stacks

Implementation & Sandbox

IMPLEMENTATION & SANDBOX

Having raw access to the reasoning stream allows systems to audit intermediate logic steps and trigger circuit breakers before dangerous tool actions execute inside a Docker sandbox.

```
Repo: github.com/deepseek-ai/DeepSeek-R1 (MIT License)
```

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